

# Choose Inputs and Outputs

Project: Helpful Classroom Sorter

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Find a Real Problem

Without opening the old lesson, explain one key idea from Find a Real Problem. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Verify Important Outputs

Without opening the old lesson, explain one key idea from Verify Important Outputs. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Privacy by Choosing Less

Without opening the old lesson, explain one key idea from Privacy by Choosing Less. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## 2. Practise with fading help

Use W2: Run the baseline on three requests for Requirements and a simple baseline. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

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## 3. Individual reasoning check

Use the worked case for Requirements and a simple baseline. Proposed explanation: “Quiet required No excludes quiet activities”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

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## 4. Independent transfer case

## Independent case: Build a baseline for a different catalogue

A paper room recommender uses R1 first, capacity 6, quiet Yes; then R2, capacity 10, quiet No. Valid requests give a whole group size from 1 to 12 and Quiet required Yes or No. A room fits when capacity is at least group size and the quiet condition is met; choose the first fitting room. Record outputs and reasons for 6 people Yes, 8 people No and 8 people Yes. Explain how a manual baseline can be compared fairly with a prototype. No real booking is allowed.

### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## 5. Your learning target

Design a paper/plastic sorter that receives an object description. Include a mixed paper-and-plastic cup. Specify inputs, outputs and the uncertain-case action. Is the owner's name useful?

## 6. Solve the given case

A sorter suggests paper or plastic trays using material observations.

Name an input, output and unnecessary input.

Show your trace, calculation or evidence

### Write or draw your reasoning and evidence.

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## 7. Check a changed case

A paper sorter must suggest pencil tray or book shelf. Choose one needed input and state the output. What should happen if the picture is blocked?

My independent answer and reason

**Write or draw your reasoning and evidence.**

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### Week 30 Test and reflect

## 8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

### Test record · My evidence

**Test record:** Changed input or condition

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Expected answer or fallback

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Observed result

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Why the result occurred

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

## 9. Project checkpoint

Freeze the sorter's v1 input list, output contract, and not-sure behavior.

The project change and evidence I will keep

**Write or draw your reasoning and evidence.**

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## 10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review \_\_\_\_\_ Next step \_\_\_\_\_

## 11. Transfer practice from the original programme

Design a paper/plastic sorter that receives an object description. Include a mixed paper-and-plastic cup. Specify inputs, outputs and the uncertain-case action. Is the owner's name useful?

**My independently reasoned answer**

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Record support used: none, hint or teacher explanation.

## 12. W1 Make a requirement checkable

Replace “suggest a good activity” with the time condition, quiet condition and tie rule used in this project. Explain why Quiet required No permits B1 as well as non-quiet activities when their times fit.

**My reasoning and evidence**

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### 13. W2 Run the baseline on three requests

Using the catalogue, record the baseline output and reason for 10 minutes with quiet required, 8 minutes with quiet not required, and 7 minutes with quiet not required. Identify which no-match result is justified and why.

**My reasoning and evidence**

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### 14. W3 Separate invalid input from no matching option

Compare a request with blank minutes and Quiet required Yes to a request with 0 minutes and Quiet required Yes. What should the baseline record for each? Explain why these are different outcomes even though neither produces an activity suggestion.

**My reasoning and evidence**

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## 15. W4 Build a baseline for a different catalogue

A paper room recommender uses R1 first, capacity 6, quiet Yes; then R2, capacity 10, quiet No. Valid requests give a whole group size from 1 to 12 and Quiet required Yes or No. A room fits when capacity is at least group size and the quiet condition is met; choose the first fitting room. Record outputs and reasons for 6 people Yes, 8 people No and 8 people Yes. Explain how a manual baseline can be compared fairly with a prototype. No real booking is allowed.

### My reasoning and evidence

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## 16. W5 Challenge an unfair comparison

Team A tests a prototype on five easy requests and the manual baseline on five different requests, then reports only the prototype's successful outputs. Identify two changes needed for a fair comparison. If both methods then meet all requirements, what can and cannot be concluded about compatibility?

### My reasoning and evidence

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## 17. Feedback, return check and learning record

### Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_

\_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 18. Pictorial case practice: Choose Inputs and Outputs

A fictional classroom chooses a storage bin for model parts.

Catalogue order: B1 capacity 5, B2 capacity 9. Valid request is a whole part count 1–12.

Return first bin with capacity at least the count; none if neither fits; clarify invalid input.

## Choose Inputs and Outputs

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Validate request

Count 6 is whole and within 1-12.



#### 2. Check first bin

B1 capacity 5 is below 6; reject it.



#### 3. Check second bin

B2 capacity 9 is at least 6; accept it.



#### 4. Baseline output

Report B2 and stop; save the comparison evidence.

### CHANGE ONE THING

Change request 6 to 5.

Explain the changed result using the matching lesson.

Trace requests 10 and blank. Why should they not receive the same explanation?

### My picture-based explanation and changed-case prediction

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Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Document sorter schema, catalogue order, acceptance rule and all output types; use a manual trace as the comparison baseline.